Detector-Assisted Annotation Audits: Capture Gaps, Count Estimation, and Review Allocation

Ananda Siva Kumar Kopalli Naga Venkata[1]

[1]Independent researcher, Troy, MI, USA; ORCID: 0009-0002-5241-9922

Corresponding author: Ananda Siva Kumar Kopalli Naga Venkata (email: anand.knvasks@gmail.com).

ABSTRACT Detector-assisted annotation review retrieves errors or estimates their total; these objectives need not favor the same audit. We compare natural omissions with synthetic controls preserving every image's target count on fixed COCO and Pascal VOC cohorts with released multi-reviewer references. On the primary detector, natural omitted-node capture is 17.5 and 16.6 percentage points below uniform controls, yet exact fixed-difference estimation risk ratios are 1.019 and 0.965. For prediction-powered active testing (PPAT), the corresponding estimated count-risk ratios are 1.222 and 1.250, showing estimator-dependent synthetic optimism. Fourteen retained estimators and one post hoc public-input allocation diagnostic expose a second distinction: PPAT reduces estimated uncapped COCO risk by 93.4% against stratified fitted-power correction, but has 17.9% higher risk than a simpler stratum-census allocation at the same 200-image budget. Original primary PPAT benefits survive that comparator. Complete reference-policy and budget results disclose further reversals. The contribution is an aligned empirical benchmark and a conditional audit-design lesson: capture, count correction, and review allocation require separate evaluation. Results concern operational reference counts and conditional simulation precision, not physical completeness or a new estimator.

INDEX TERMS Annotation auditing, finite-population estimation, object detection, prediction assistance, reproducibility.

I. Introduction

DETECTING which annotations are missing and estimating how many are missing are different tasks. An unmatched detector proposal can direct a reviewer toward an error, but it can also reflect a spurious detection, localization disagreement, or annotation convention. Conversely, a detector can miss an unannotated object yet retain information about which images contain many omissions. The first task is commonly evaluated through instance retrieval. The second requires a defined total, a probability-sampling design, and accounting for every reviewed image.

Recent multi-reviewer benchmarks establish that synthetic annotation corruption can substantially overstate real-error retrieval performance [1]. This motivates, but does not answer, a different question: does the identity of naturally non-captured instances degrade *finite-budget count estimation*, after holding each image's omission count fixed? A retrieval gap alone cannot determine this. For a fixed prediction-assisted total estimator, error depends on residual variance, including both predictor variance and its covariance with the target [2], [3]. For adaptive sampling, acquisition and correction also matter [4], [5].

We address this question using released multi-reviewer references rather than assuming that constructed deletions represent natural errors. A count-matched design retains the same outcome vector across actual annotations, normalized natural retention, uniform deletions, and height-weighted deletions. This prevents altered error prevalence or image-level concentration from explaining the controlled differences. It intentionally leaves those factors outside the controlled question; it is not a general validation of synthetic benchmarks.

The study was designed to detect a material natural-error penalty. That directional hypothesis fails. We retain its threshold, models and unfavorable results. The revised evidence aligns instance capture with the exact same natural/synthetic controls, and tests an omitted simple allocation against the fixed active procedure. Both steps sharpen the scientific lesson without requiring PPAT to remain the preferred method everywhere.

The contributions are three empirical results. First, aligned count-matched controls quantify a natural capture disadvantage that translates differently across count estimators and detectors; fixed-predictor moments explain the primary near-

cancellation. Second, a same-budget public-input allocation exposes a ranking reversal hidden by a large named-baseline improvement. Third, complete policy/budget comparisons and an executable selected-image measurement interface delimit what this evidence can recommend. Established estimators, the source reference and classical variance identities are credited; neither a new algorithm nor a generally optimal audit policy is claimed.

II. Closest Work and the Distinction Tested

Penquitt et al. contribute multi-reviewer references, uncertainty-aware detection evaluation, and real-error retrieval benchmarks [1]. Our references are derived from their release, with explicit exclusions and policies. We do not contribute their annotations or rediscover their natural/synthetic retrieval gap. Our endpoint is squared error in estimating an operational total at a fixed reviewed-image budget. The controlled contrast fixes the target vector, unlike comparisons that also alter error prevalence or image difficulty.

ObjectLab and loss-inspection methods already use model information to identify object-detection annotation errors [6], [7]. Unmatched proposal counts are therefore not a novel auditing idea. SALUS already uses manually checked samples and calibration to estimate benchmark-wide annotation-error prevalence in another domain [8]. An annotation-error total or calibration step alone would not distinguish this work.

Model-assisted survey estimation, prediction-powered inference (PPI), Cross-PPI, stratified PPI, and PPI++ establish prediction-assisted correction and related efficiency ideas [2], [3], [9]–[12]. Mani et al. analyze finite-sample risks of mean-estimation PPI++ and show that prediction assistance need not improve risk [13]. Generic finite-sample failure is therefore not our novelty claim; we study matched natural omissions and the effect of review allocation on count-risk comparisons.

Active testing, LURE, and PPAT establish adaptive acquisition with corrected estimation [4], [5], [14]. We compare disclosed count specializations with fixed features and tuning, rather than claiming complete native reproductions or new general algorithms. ATLAS already couples adaptive categorical-label auditing to confidence sequences and task-dependent review costs [15]; our study does not establish superiority over its full framework.

The distinct empirical question is whether an instance-retrieval disadvantage translates into a count-estimation penalty when per-image counts are matched, and how review allocation changes the practical ranking under actual annotations and explicit reference policies. Penquitt et al. do not claim that poor retrieval makes aggregate estimation impossible; our study evaluates a different endpoint rather than refuting such an assertion. This is a conditional diagnostic of benchmark interpretation. Neither the general variance identity nor the statement that predictions can help despite errors is new. The contribution depends on the measured size, replication, strong comparisons, and reference limitations of the specific result.

TABLE 1. Operational reference counts in each 1,000-image frame. "Cap" counts affected images.

Frame	Policy	Refs.	Uncapped	Target	Cap
VOC train	I	902	140	140	0
VOC train	S	780	73	73	0
COCO	I	2324	733	600	4
COCO	S	1758	394	365	3
VOC val	I	1022	177	177	0
VOC val	S	891	95	95	0

III. Reference, Target, and Controlled States

A. Source and eligibility

We use three fixed 1,000-image cohorts: VOC 2012 train for development, COCO val 2017 and VOC 2012 validation for confirmation [16], [17]. Cohorts were fixed before natural-audit performance was examined. Image identities, dimensions, official splits, and all 3,000 image hashes were checked; there are no exact image-hash duplicates across the cohorts. COCO synthetic results from an earlier study were already known. Confirmation here means a fixed analysis stream, not an entirely unfamiliar dataset or an independently collected reference.

The reference is the pinned soft-label release of Penquitt et al. [1]. We exclude records marked as original-seeded additions. This separates discovery provenance; it does not assert that excluded records lacked human review. The source describes review and class voting for accepted original-only additions. Eligible boxes have positive area, argmax class person, and satisfy one of two frozen policies: inclusive (I), class score > 0.5 and height at least 25 pixels; strict (S), score > 0.8 and height at least 40 pixels. These scores express released class judgments, not probabilities that an object physically exists or that its box is spatially exact.

Original person annotations are matched to eligible reference boxes at intersection-over-union (IoU) at least 0.5, maximizing cardinality and then summed IoU. VOC inclusive one-based coordinates are converted to zero-based half-open boxes; COCO width/height boxes are converted and clipped to the image. Original crowd/difficult records remain possible captures. Stable ordering makes the implementation deterministic; an equally optimal matching can assign different individual nodes.

For reference set R_i and captured subset C_i, write $m_i = |R_i| - |C_i|$. The measured target is

$$Y_i = \min(20, m_i), \qquad B = \sum_{i=1}^{N} Y_i, \quad N = 1,000. \tag{1}$$

This is a capped count of *reference noncaptures*. It is not the number of all physical omissions or necessarily the number of violations of an original dataset's policy. The source

itself can miss objects. Table 1 discloses the cap's effect: on inclusive COCO, 733 uncapped noncaptures become a target of 600. Four images are capped, including one with 115 noncaptures. No VOC confirmation image is capped.

B. Four states with identical targets

State A uses the actual original annotations and their geometry. State N retains exactly the captured reference nodes C_i, at reference coordinates. State U retains a uniformly chosen subset of R_i of size $|C_i|$. State H retains the same number after weighted deletion, with weights 0.50,0.20,0.05 for heights below 40,40–80, and at least 80 pixels. Hash-seeded exponential races implement the fixed weighted selection without replacement.

Every retained node in N/U/H has an identity match to the full reference. Therefore the maximum matching has exactly the retained-set cardinality, and all four states have the same uncapped m_i and capped Y_i. This remains true for overlapping boxes because reference nodes retain distinct identities. Direct sampling using a shared image sample must consequently give exactly the same estimate in every state; execution checks this path by path.

N versus U/H changes which instances are omitted conditional on each image's count. It does not test changes in the distribution of counts across images. A versus N changes original geometry *and* the annotation-policy pool, so it is not a pure localization contrast. Constructing U/H requires the full natural counts and reference boxes. These are retrospective diagnostic controls, not a label-free deployment procedure.

IV. Detectors, Estimators, and Fixed Analysis

A. Detectors and public information

We freeze COCO-pretrained Faster R-CNN ResNet-50-FPN-v2 and RetinaNet ResNet-50-FPN-v2 [18]–[20]. There is no detector training or threshold search. TorchVision 0.23.0 with Torch 2.8.0 uses float32 inference, default 800/1333 resizing, NMS 0.5, and a maximum of 100 post-NMS outputs across classes. The RetinaNet output cap is explicitly set to 100. We keep person predictions with score at least 0.5 and height at least 25 pixels. One thousand earlier Faster R-CNN COCO predictions are reused; 5,000 new predictions and ten retained replays are hash-verified.

Public image features are unmatched proposal count, annotation count of height at least 25, detector count, and unmatched counts in three height bands. The offset model adds three leave-one-image-out full-frame means. With one bookkeeping group these are affine copies of existing count features; they add no independent context, although they affect ridge regularization. The scalar proxy is $q_i = \min(20, \text{unmatched proposals})$. Actual-state features use only detector outputs and original annotations. Normalized and synthetic features incorporate the reference by construction, as required by the conditional comparison.

B. Budget and comparison set

Budgets $b \in \{50, 100, 200\}$ count distinct reviewed images, including every pilot and training image. Empty images also cost one review. Reference construction, detector pretraining, full-frame inference, and engineering are outside this budget. Equal image budgets do not establish equal human time.

The fourteen estimators are direct and stratified Horvitz–Thompson (HT); fixed difference; half difference; split ridge; split offset; sequential offset; fitted power; stratified fitted power; cross-fit offset; cross-fit direct; active LURE count; and PPAT count with fixed coefficients 0.5 and 1. Their exact definitions appear in the supplement. The stratified methods use proposal-count strata 0,1, and at least 2 with deterministic exact allocation. Split methods spend $\lfloor b/5 \rfloor$ reviews on fitting. Sequential and active models refit using only completed observations at 20,40,80,120, and 160 reviews when reached. Cross-fitting uses two fixed equal frame folds, charging both samples once.

For fixed scalar c, the difference estimator is

$$\widehat{B}_c = \frac{N}{b} \sum_{i \in J} (Y_i - c q_i) + c \sum_{i=1}^{N} q_i, \tag{2}$$

where J is a simple random sample without replacement. It is unbiased for fixed q, c. Fitted-power coefficients are estimated from the same acquired labels and clipped to $[0, 1]$; exact unbiasedness is not asserted for those fitted estimators.

For the active count specializations, set $c_i = q_i - \bar{q}$ and residual $r_i = Y_i - \lambda c_i$. A past-trained count forecast $\mu_{t,i} \in [0, 20]$ gives acquisition scores

$$a_{t,i} = \sqrt{\mu_{t,i}(1 - \mu_{t,i}/20) + (\mu_{t,i} - \lambda c_i)^2}. \tag{3}$$

The proposal mixes 90% normalized scores with 10% uniform probability on remaining images; a zero score sum gives uniform sampling. LURE's finite-population correction estimates the residual total, which equals B because $\sum_i c_i = 0$. Coefficients 0,0.5, and 1 give the three active comparators. This surrogate and feature specification is a count adaptation of existing methods, not their complete original experimental pipeline.

C. Frozen repetitions, selection, and inference

Each confirmation frame/policy/detector cell uses 50 deletion blocks and 20 independent audit paths per block, for 1,000 evaluation paths. Development uses 10 blocks. A/N are fixed; U/H each receive one deletion realization per block. A separate 10-block,200-path synthetic stream selects the method with lowest pooled U/H MSE at $b = 200$, with ties resolved by the fixed method order. Evaluation never chooses the method using its own errors. The choice remains conditional on this retrospective selection experiment; it is not free or guaranteed to transfer prospectively.

States and methods share random streams. Active acquisition uses common uniforms, not forced identical image samples. The study records MSE, bias, median and 90th-percentile absolute error, and label counts. All methods,

budgets, policies, and detectors remain in the complete tables. Twelve primary contrasts use I/Faster/$b = 200$, six per confirmation frame: N/U risk ratios for fixed difference and sequential correction, and selected-method risk versus stratified HT and stratified fitted power in A and N.

The original analysis used 10,000 paired block-bootstrap resamples. For this revision, two independent streams of one million resamples per cell are pooled, retaining the 20 paths within each block. Both streams must agree in directional classification and have endpoints within 2% of the pooled limits; all 84 primary, secondary and post hoc contrasts pass this fixed precision check. More resamples improve numerical tail precision, not the number of independent blocks or statistical coverage. Two-sided percentile limits allocate family $\alpha = 0.05$ across the 12 contrasts. These approximate limits describe Monte Carlo precision conditional on the finite frames, fixed controls, and selected method; they are not uncertainty about physical truth or all future datasets. The prespecified material-harm criterion requires a ratio at least 1.5 and adjusted lower limit above 1 in the same contrast on both confirmation frames. The 1.5 threshold is an author-defined practical criterion, not a journal rule. The opposite-direction interpretation below is reported transparently; the failed harm criterion is not redefined as passed.

V. Results

A. The prespecified material-harm gate fails

None of the primary harm contrasts meets the declared criterion. Fixed-difference N/U MSE ratios are 0.976 on COCO and 0.956 on VOC; sequential ratios are 1.051 and 1.022. Their adjusted upper Monte Carlo limits are below 1.14 in both frames. This provides evidence against a penalty as large as the declared 1.5 threshold in these specific contrasts, but it is not a formal equivalence claim or a guarantee about unmatched count distributions outside the design.

The separate synthetic selector chooses PPAT count 0.5 in both primary cells. In actual annotations, its MSE is 1,223.10 on COCO and 507.42 on VOC, corresponding to root MSE 34.97 and 22.53 counts. Relative to stratified fitted power, the primary ratios are 0.337 and 0.524: observed MSE reductions of 66.3% and 47.6%. Adjusted upper limits remain below 1. Relative to stratified HT the ratios are 0.104 and 0.440. All fitting labels are included in the 200-image budgets.

The selected method has the lowest observed primary MSE among all fourteen implementations (Fig. 1). This descriptive ranking is not a simultaneous claim of superiority over every method. Its actual-state biases are 0.075 and 0.086 counts, with 90th-percentile absolute errors 57.10 and 35.98. Direct HT root MSE is 138.13 and 41.31. These are simulated review-design errors for fixed operational totals 600 and 177, not observed errors from independent human audits.

B. Aligned capture and count risk show estimator dependence

The detector-information comparison now uses exactly the same omitted-node sets as the estimation contrast. For each fixed full-reference detector matching, we measure the fraction of omitted nodes that it captures. Natural N and synthetic U/H have the same uncapped omitted-node denominator within each frame/policy. Detector capture is an uncapped instance-level quantity; the primary count target is capped per image. We therefore also recompute the actual capped public proxy q and its residual risk for every saved control. Equally optimal matchings can assign different nodes, so this diagnostic is operational rather than a causal effect or a retrieval benchmark replication.

On primary COCO, N capture is 354/733 (48.3%), versus mean U capture 65.8%; on VOC it is 105/177 (59.3%), versus 75.9%. The matched gaps are 17.5 and 16.6 percentage points. Mean H capture is 61.2% and 71.8%, respectively. Reconstructing all eight cells verifies 800,000 saved feature rows and 800 deletion blocks. The potentially identity-changing images carry 568/600 of COCO's capped target and 137/177 of VOC's target, so the controls do not merely perturb target-free images. Compared with N, U changes capped q on a mean 82.24 and 28.58 images, respectively.

For fixed q, c, the established exact design-risk identity is

$$\mathrm{MSE}(\widehat{B}_c) = \frac{N^2(1-b/N)}{b}\left(S_Y^2 + c^2 S_q^2 - 2cS_{Yq}\right), \quad (4)$$

with denominator $N-1$ in the finite-frame moments. Because Y is identical across N/U, the fixed-difference risk contrast is determined by $\Delta S_q^2 - 2\Delta S_{Yq}$. On COCO, moving from mean U moments to N reduces predictor variance by 1.035 and covariance by 0.528; these nearly cancel in residual variance. Exact N/mean-U fixed-difference risk is 1.019. On VOC, the corresponding reductions 0.143 and 0.058 yield a ratio 0.965. These exact risks have no audit-simulation error; the saved Monte Carlo ratios in Table 2 fluctuate around them. The fixed 50 U/H realizations still condition this interpretation.

This does not make natural and synthetic errors interchangeable. On the same primary controls, saved PPAT N/U MSE ratios are 1.222 and 1.250. Inclusive RetinaNet has capture gaps of 21.7 and 28.7 percentage points, with exact fixed-difference risk ratios 1.590 and 1.184 and PPAT ratios 1.442 and 1.795. Thus lower natural capture coexists with near-unchanged fixed-difference risk for Faster R-CNN, but meaningful synthetic optimism for other estimator/detector choices. Complete N/U and N/H capture, proxy and risk decompositions appear in the supplement.

The distinct empirical result is this aligned, endpoint- and estimator-dependent response at unchanged image-level targets. It does not claim that the general possibility of retrieval/estimation divergence is new. Complete-frame moments explain the observed cases but are unavailable before review, and Eq. (4) cannot be applied to fitted or adaptive estimators by simple substitution. The practical lesson is to

TABLE 2. All prespecified primary MSE ratios at $b = 200$, inclusive reference, Faster R-CNN. Brackets are approximate block-bootstrap limits adjusted across twelve contrasts, updated with two million resamples. The original material-harm criterion remains unmet.

Contrast	COCO ratio [limits]	VOC ratio [limits]
N/U: fixed difference	0.976 [0.905, 1.053]	0.956 [0.908, 1.007]
N/U: sequential	1.051 [0.974, 1.139]	1.022 [0.974, 1.070]
A: selected/stratified HT	0.104 [0.087, 0.124]	0.440 [0.370, 0.521]
A: selected/stratified power	0.337 [0.286, 0.395]	0.524 [0.437, 0.635]
N: selected/stratified HT	0.104 [0.090, 0.121]	0.530 [0.443, 0.630]
N: selected/stratified power	0.353 [0.302, 0.412]	0.652 [0.546, 0.775]

assess the intended count estimator and allocation directly rather than treating instance capture as a sufficient criterion for count-audit suitability.

C. Prespecified robustness and reference sensitivity

At the primary budget, the selected method has lower observed actual-state MSE than stratified fitted power in 7 of the eight fixed cells, with ratios from 0.147 to 1.000 (Table 3). Across all three budgets this occurs in 20 of twenty-four cells. Depending on the cell, the selector chooses either stratified fitted power or PPAT count (0.5). These secondary comparisons delimit the result; their descriptive rankings are not added to the primary family after observing outcomes. All state and method results, including reversals, appear in the supplement and machine-readable tables. In strict VOC with Faster R-CNN, the selector chooses stratified fitted power itself; its actual-state MSE is 6.1% higher than stratified HT at 200 reviews. At 50 reviews, inclusive VOC with RetinaNet has 4.6% higher selected-method MSE than stratified fitted power. Thus the selection procedure does not uniformly improve on every simpler comparator. The natural/synthetic effect also depends on architecture. For inclusive COCO with RetinaNet, fixed-difference N/U MSE is 1.590, while the corresponding VOC ratio is 1.169. The COCO secondary cell crosses the material-harm point threshold, but the effect does not meet the same threshold in both frames and is not promoted to a successful primary result. This rules out a blanket claim that omission identity never matters. The selected active method itself has descriptive N/U MSE ratios of 1.222 on COCO and 1.250 on VOC at the primary budget. Thus synthetic optimism is not absent for every estimator, even though the declared primary harm gate fails and actual-state estimation remains useful. These selected-method N/U ratios are secondary, not additional adjusted primary tests. Policy and cap changes alter the operational target, so results across I/S cannot be treated as repeated measurements of one physical truth. In inclusive COCO, 347 of 733 unmatched reference centers fall within crowd/difficult original boxes; the corresponding VOC count is 16/177. Two COCO and 93 VOC reference boxes are matched to crowd/difficult originals. These are reported diagnostics, not automatic proof that the remaining boxes are errors or that ignored-region overlaps should be deleted after seeing results.

VI. A Fixed Audit Procedure and Reference Sensitivities

A. Executing the audit without a full-frame reference

The synthetic selector is a retrospective diagnostic: it uses the known natural count vector to construct controls and cannot be assumed available to an auditor. A separate operational procedure fixes PPAT count $1/2$ before sampling. The development review recorded this choice in all four development cells before confirmation performance was inspected, and $1/2$ is already a published PPAT heuristic [5]. Reporting it uniformly now is a secondary analysis, not a newly prespecified primary success.

An auditor needs an image inventory, original annotations, frozen detector outputs, a declared reference policy and a callback that measures only the selected image. No full-reference outcome vector, synthetic deletion set or confirmation-frame method search is required. For each image, compute the public count features described above and $q_i = \min(K, \text{unmatched predictions}_i)$; let $c_i = q_i - \bar{q}$. With $M_t = N - t + 1$ images remaining before draw t, sample according to

$$p_t(i) = 0.9\frac{a_t(i)}{\sum_{j\in R_t} a_t(j)} + \frac{0.1}{M_t},$$
$$a_t(i) = \sqrt{\mu_t(i)(1-\mu_t(i)/K) + (\mu_t(i) - c_i/2)^2}. \tag{5}$$

If all scores vanish, use uniform sampling. Forecasts start at q and use the retained offset-ridge update with penalty 100 after 20,40,80,120,160 observed images. Review the whole selected eligible image, rather than only detector proposals, to obtain Y_{i_t}. With $b < N$, the established LURE-corrected point is

$$\widehat{B} = \sum_{t=1}^{b} \frac{N}{b}\left[1 + \frac{N-b}{N-t}\left(\frac{1}{M_t p_t(i_t)} - 1\right)\right]\left(Y_{i_t} - \frac{c_{i_t}}{2}\right). \tag{6}$$

The centered public proxy sums to zero. These formulas specify the existing implementation; they are not new statistical results. Every acquired image, including all fitting observations, counts once in b. The accompanying callback interface reproduces the saved points and complete traces in all eight original confirmation cells. Altering unobserved outcomes leaves the tested paths unchanged.

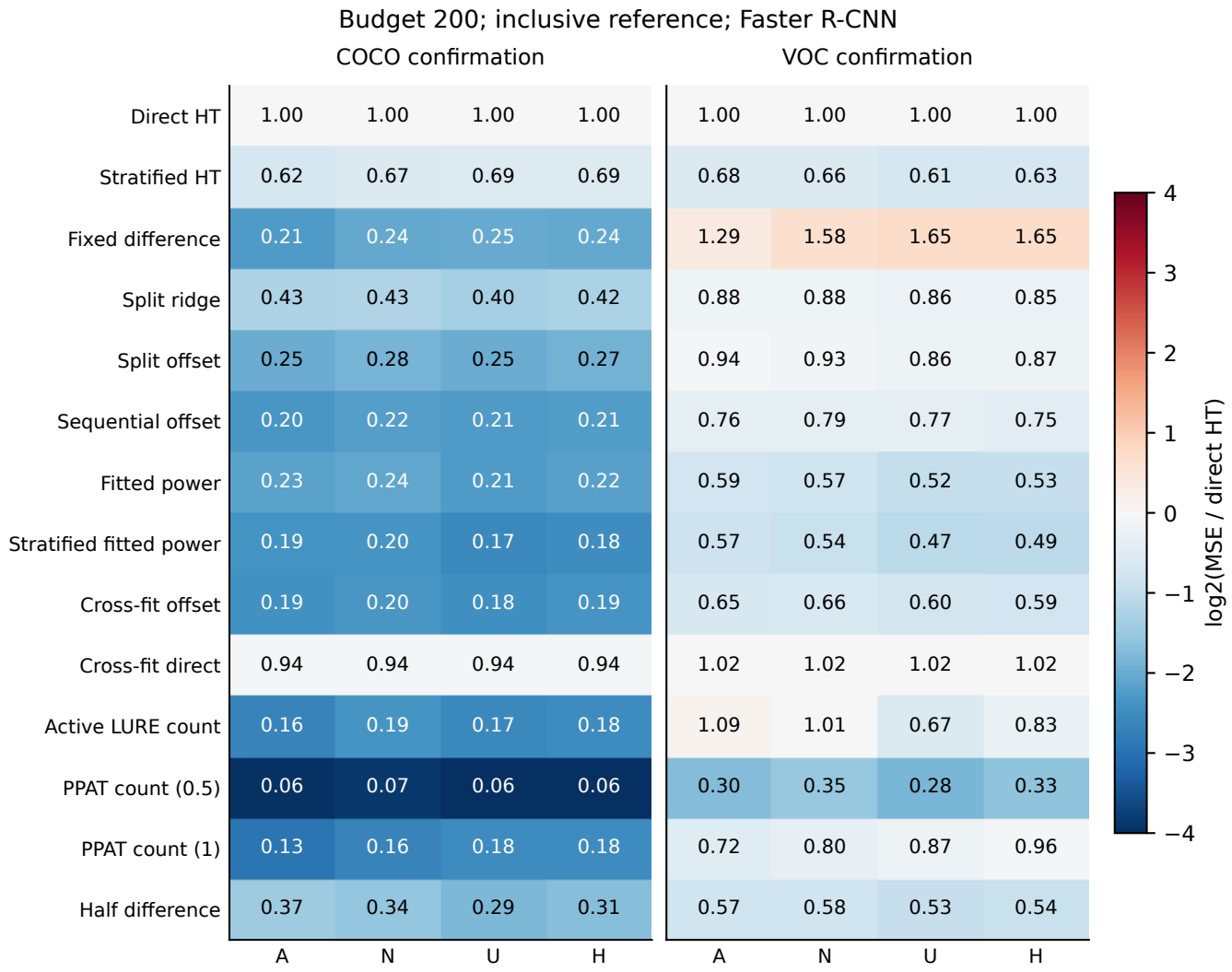


FIGURE 1. **All fourteen estimators at the primary budget and policy. Entries are MSE divided by direct HT MSE within each confirmation frame; colors clip at 1/16 and 16. The direct estimate is pathwise identical across states. A uses actual annotations; N normalizes naturally captured reference nodes; U/H change deleted-instance identity at the same image-level counts. Method names denote the disclosed implementations, not full native pipelines.**

Reference quality remains a measurement requirement. Replacing the released multi-reviewer judgments with an unvalidated single reviewer would change the evidence basis. The budget counts distinct reviewed images, not rater-hours, and excludes detector inference and reference creation. The procedure returns a fixed-budget point estimate; this study supplies no natural-reference confidence certificate or validated optional-stopping rule.

B. Secondary sensitivity design

Following review of the original findings, we froze two additional reference conditions before their performance calculations. In *uncapped*, outcomes are the original uncapped noncapture counts. The implementation bound is fixed to $K = 200$ and all observed counts are at most 115, so no outcome is truncated in these cohorts. This bound also changes proxy clipping, forecast clipping and acquisition; its comparison with $K = 20$ is not a pure intervention on outcome truncation alone.

In *ignore*, original crowd/difficult boxes define masks. We exclude reference boxes and predictions centered in a mask; remove original crowd/difficult boxes and other originals centered in a mask; then rematch the remaining originals and references. Mask boundaries are inclusive. Outcomes remain capped at 20. This changes the operational target and public features together. An excluded object is not thereby judged invalid. Both conditions reuse the fixed detector outputs and retain all eight confirmation cells.

At $b = 200$, each condition compares PPAT count $1/2$ with direct HT, proportional-stratified HT and stratified fitted-power correction using the original 50 blocks of 20 stream-indexed paths. Only the two new conditions are simulated; original outcomes are reused. The 800 new blocks contain 64,000 estimates. A new 48-contrast secondary family covers the fixed method versus both stratified comparators



FIGURE 2. **Capture among omitted reference nodes in the same N/U/H controls used for estimation. Detector-to-full-reference maximum matching is fixed; N is natural noncapture, U/H are the 50 saved count-matched deletion sets. Whiskers show the minimum and maximum across those fixed controls, not confidence limits. This node-capture diagnostic is not error-retrieval precision/recall. I/S denote inclusive/strict policies.**

TABLE 3. **Fixed robustness cells at $b = 200$. Ratios concern the independently synthetic-selected method in actual annotations. Denominators are direct HT, stratified HT, and stratified fitted power. Only I/Faster rows belong to the primary family; other rows are prespecified secondary evidence.**

Frame	Policy	Detector	Selector	RMSE	/Direct	/Strat. HT	/Strat. power
COCO	I	Faster	PPAT 0.5	34.97	0.064	0.104	0.337
COCO	I	Retina	PPAT 0.5	52.17	0.131	0.246	0.261
COCO	S	Faster	PPAT 0.5	36.84	0.121	0.161	0.445
COCO	S	Retina	PPAT 0.5	31.93	0.091	0.137	0.147
VOC val	I	Faster	PPAT 0.5	22.53	0.297	0.440	0.524
VOC val	I	Retina	PPAT 0.5	26.58	0.418	0.682	0.667
VOC val	S	Faster	Strat. power	20.68	0.742	1.061	1.000
VOC val	S	Retina	PPAT 0.5	15.89	0.457	0.642	0.626

across three reference conditions and eight cells, with the same two-million-resample precision update and Bonferroni allocation of $\alpha = 0.05$. The limits measure approximate Monte Carlo precision conditional on the frames. The original 12 primary contrasts and failed material-harm criterion are unchanged.

C. Named-baseline gains and a policy reversal

The same fixed method has lower observed original-state MSE than both specified stratified comparators in all eight $b = 200$ cells. This ranking is descriptive: in strict VOC/Faster, ratios of 0.953 to stratified HT and 0.899 to stratified power have adjusted limits crossing 1. Smaller-budget failures are substantial. At $b = 50$ in that cell, the method has 49.0% higher MSE than stratified HT and 25.6% higher than stratified power; at $b = 100$ it remains 7.8% worse than stratified HT. Thus 200 is an evaluated budget on these 1,000-image frames, not a validated universal sample-size recommendation. All eight required secondary sensitivity contrasts pass the frozen robustness criterion: in both I/Faster frames and both new reference conditions, the fixed method improves on both stratified comparators with adjusted upper limits below 1 (Table 4). Against stratified fitted power, reductions are 93.4% and 43.2% for uncapped COCO and VOC, and 56.9% and 41.3% for the ignore condition. These are within-condition method comparisons, not equal-target comparisons across policies. The broader family prevents an unrestricted recommendation. Forty of 48 comparisons support improvement; six are inconclusive by the adjusted limits; two support worsening. Both worsening contrasts occur for strict COCO/RetinaNet under ignore: ratios are 1.625 [1.156, 2.225] to stratified HT and 1.589 [1.131, 2.171] to stratified power. Its original-state ratio to stratified power was 0.147. Because the exclusion rule changes both outcomes and public predictors, this is a policy-conditional reversal, not an isolated causal effect of removing a particular object. Inclusive COCO/RetinaNet also has worse observed ratios under ignore, but its adjusted limits cross 1. These gains are relative to the two named, proportional-allocation baselines. They do not establish the preferred allocation or count estimator. The additional comparison below changes the practical interpretation of the large uncapped gain.

D. A public-input allocation check changes the recommendation

The original stratified baselines allocate proportionally. A post hoc diagnostic instead censuses the existing at-least-two-proposals stratum when it fits within $b = 200$ after reserving $\min(N_h, 2)$ reviews for each other stratum, and

TABLE 4. Fixed PPAT count $1/2$ at 200 reviews on the primary I/Faster frames. Ratios compare its MSE to stratified HT and stratified fitted power; brackets are adjusted Monte Carlo limits from the new 48-contrast secondary family. Reference conditions define different targets.

Reference	Frame	Target	RMSE	/Stratified HT	/Stratified power
Original	COCO	600	34.97	0.104 [0.085, 0.127]	0.337 [0.279, 0.403]
Original	VOC	177	22.53	0.440 [0.361, 0.533]	0.524 [0.426, 0.653]
Uncapped	COCO	733	49.80	0.037 [0.031, 0.045]	0.066 [0.055, 0.079]
Uncapped	VOC	177	23.46	0.477 [0.399, 0.567]	0.568 [0.464, 0.689]
Ignore	COCO	315	33.71	0.304 [0.245, 0.382]	0.431 [0.354, 0.531]
Ignore	VOC	161	22.38	0.482 [0.385, 0.602]	0.587 [0.468, 0.747]

distributes the remainder proportionally over their remaining capacities with largest-remainder rounding. If it does not fit, the original allocation is used. The design uses only public stratum sizes and proposal counts; outcomes enter only its evaluation. The rule was chosen during review after the original results were available, so its selection is not prospective. No threshold or rule search was performed.

For fixed stratum samples, the established stratified HT risk is $\sum_h N_h^2(1 - n_h/N_h)S_{Y,h}^2/n_h$, with zero contribution from censused strata. It is evaluated exactly here, rather than estimated from new simulations. Both methods spend exactly 200 distinct-image reviews; the allocation rule needs no fitted model or free pilot labels.

For COCO/Faster the stratum sizes are 750,126,124. Proportional allocation is 148,26,26, whereas the diagnostic uses 64,12,124. On uncapped COCO, its exact MSE is 2,103.83 versus PPAT's 2,479.67: the simpler rule has 15.2% lower risk by the point comparison, despite PPAT's 93.4% gain over the named fitted-power baseline. The PPAT/allocation ratio is 1.179 [1.016,1.345]. The high-proposal stratum contains all four capped images, so censusing it also makes this comparator's original and uncapped risks equal. This is a concrete allocation effect, not evidence that clipping never matters.

PPAT retains lower MSE on both original primary frames (ratios 0.581 and 0.798). Across all 24 conditions, the diagnostic has lower point risk in eight; the adjusted limits support seven of these, leave one inconclusive, and support PPAT in fourteen of the other sixteen. Two further comparisons are inconclusive. The supplement retains all conditions. These limits measure only simulation precision for the fixed post hoc rule.

The fitted-power implementation clips coefficients to $[0, 1]$. On uncapped COCO/Faster, the full-reference coefficient in the high-proposal stratum is 1.846, outside that interval. This diagnostic explains a potentially consequential restriction on a count scale; it is not an oracle deployable baseline or a measured result for an unrestricted fitted estimator. PPI++ discusses constrained and one-step alternatives [12]. The present result cannot establish superiority over all calibration choices.

Thus the surviving recommendation is conditional: PPAT-half is a supported candidate in the original I/Faster/$b = 200$ setting, but the large uncapped named-baseline improvement does not justify preferring it over a simple allocation. Auditors should compare sampling allocations as well as correction methods for their declared target. Neither the retrospective selector nor this post hoc rule provides a validated way to choose the best method on an unseen frame.

VII. Limitations and Practical Implications

The original controlled result fixes the per-image target, deliberately removing image-level error concentration as an explanation. It therefore does not validate arbitrary synthetic benchmarks or test concentration shifts. The later fixed-procedure and reference-sensitivity analyses are explicitly secondary. They do not convert the failed original directional hypothesis into a prospective success.

References share one source pipeline, and class scores do not model localization uncertainty. Both original and released annotations can miss the same object. Against the original proportional-allocation baselines, the primary benefits survive the two separate reference sensitivities; the new allocation comparator reverses the uncapped COCO ranking. These analyses do not cover every policy, their joint combination, or physical truth. The uncapped condition changes bound-dependent acquisition as well as the target. Bootstrap limits quantify conditional simulation precision and do not propagate uncertainty about reference completeness or new human measurement.

Only one class, two related detector architectures, two confirmation datasets and the specified methods are evaluated. Training-corpus overlap has not been exhaustively audited, so unseen-domain generalization is not claimed. The original stratified baselines use proportional allocation; the additional stratum-census diagnostic is post hoc and is neither an exhaustive allocation comparison nor prospectively validated. Unrestricted fitted calibration and full adaptive Neyman allocation remain unevaluated; count adaptations are not full native reproductions of every modern auditing pipeline. The same-sample fitted-power estimator may be biased. We report its bias and do not claim that the strongest observed method is universally optimal. Retrospective synthetic selection remains separate from the executable fixed

TABLE 5. Post hoc allocation diagnostic, primary I/Faster frames. The numerator is saved PPAT-half MSE; the denominator is exact stratified HT risk for the public-input allocation. Brackets are approximate conditional Monte Carlo limits from the new 24-contrast family, not prospective validation of the rule.

Condition	Frame	Allocation RMSE	PPAT RMSE	PPAT/allocation MSE
Original	COCO	45.87	34.97	0.581 [0.510, 0.658]
Original	VOC	25.22	22.53	0.798 [0.695, 0.929]
Uncapped	COCO	45.87	49.80	1.179 [1.016, 1.345]
Uncapped	VOC	25.22	23.46	0.865 [0.747, 0.988]
Ignore	COCO	41.94	33.71	0.646 [0.558, 0.746]
Ignore	VOC	22.66	22.38	0.976 [0.848, 1.120]

procedure, whose post-result presentation does not establish prospective selection validity.

An auditor should define the measurement policy and budget first, review selected images completely, and report corrected total estimation separately from error retrieval. On the original I/Faster frames at $b = 200$, PPAT-half improves on the reported alternatives, including the added allocation. That recommendation does not extend uniformly to the uncapped, low-budget or RetinaNet policy conditions. Its root MSE is 34.97/600 (5.8%) and 22.53/177 (12.7%) of the operational target. Those errors could inform the scale of an annotation-review effort, but no application-specific tolerance or downstream decision benefit was measured. A deployment decision still requires a tolerance, an appropriate measurement process and a justified sampling allocation. Human time, downstream detector retraining gains and natural-reference interval coverage were not measured. Creating an accurate total estimate is not equivalent to repairing the annotation set.

VIII. Conclusion

Aligned natural/synthetic controls reveal substantial capture gaps but estimator-dependent count-risk changes at identical image-level targets. Exact moments explain the primary fixed-difference near-cancellation, while PPAT and RetinaNet retain synthetic optimism. A same-budget allocation comparison also reverses a headline uncapped ranking. The supported contribution is the measured separation of capture, correction and allocation, with explicit policy/budget limits and reproducible evidence. It supports a conditional audit-design recommendation, not universal superiority or physical-completeness certification.

Availability and Declarations

The author reports no external funding and no competing interests. Reproducibility materials for this submission are deposited in Zenodo under doi:10.5281/zenodo.24126429, with a code and release mirror at https://github.com/anandakopalli7/programme-b-annotation-audits. The current Supplementary Evidence archive (submission-v1) is also supplied with this manuscript and supplement. Its entry point identifies current source, protocols, hashes, detector predictions, raw blocks, allocation and aligned-control results, and resampling records. It is self-contained for numerical verification. Dataset images and model weights are excluded. Adapted reference data retain the source's CC BY-SA 4.0 attribution and share-alike terms; original code retains its separate license.

Acknowledgment

The author directed the study and set its objectives. Under the author's instructions, OpenAI Codex assisted with literature searches, mathematical derivations and checks, study implementation, analysis, numerical verification, and manuscript preparation. Text in the abstract, Sections I–VIII, the figure and table captions, and the supplementary material was drafted and revised with Codex. The supporting mathematical arguments, analysis and plotting code, and reproducibility package were developed with Codex assistance. Reported numerical results are computed from retained experimental records. The author is responsible for the scientific claims, citations, analysis, and final manuscript.

REFERENCES

[1] S. Penquitt et al., "Object detection benchmarks are incomplete: The role of label errors and annotation uncertainty," 2026, arXiv:2609.21822. [Online]. Available: https://arxiv.org/abs/2609.21822.

[2] C.-E. Särndal, B. Swensson, and J. Wretman, *Model Assisted Survey Sampling*, Springer Series in Statistics. Springer, 1992. [Online]. Available: https://link.springer.com/book/9780387406206.

[3] F. J. Breidt and J. D. Opsomer, "Model-assisted survey estimation with modern prediction techniques," *Statistical Science*, vol. 32, no. 2, pp. 190–205, 2017, doi: 10.1214/16-STS589.

[4] S. Farquhar, Y. Gal, and T. Rainforth, "On statistical bias in active learning: How and when to fix it," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2021. [Online]. Available: https://arxiv.org/abs/2101.11665v2.

[5] K. Ashouritaklimi, V. Kilian, D. Huang, T. Rainforth, and F. Caron, "Prediction-powered active testing," 2026, arXiv:2607.08347v1. [Online]. Available: https://arxiv.org/abs/2607.08347v1.

[6] U. Tkachenko, A. Thyagarajan, and J. Mueller, "ObjectLab: Automated diagnosis of mislabeled images in object detection data," 2023, arXiv:2309.00832. [Online]. Available: https://arxiv.org/abs/2309.00832.

[7] M. Schubert et al., "Identifying label errors in object detection datasets by loss inspection," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, 2024, pp. 4582–4591. [Online]. Available: https://arxiv.org/abs/2303.06999.

[8] S. Zhou, A. G. Colaco, S. Galhotra, and S. Mehrotra, "SALUS: Automated auditing of NL-to-SQL benchmarks through weak supervision of multi-agent output," 2026, arXiv:2610.05540. [Online]. Available: https://arxiv.org/abs/2610.05540.

[9] A. N. Angelopoulos, S. Bates, C. Fannjiang, M. I. Jordan, and T. Zrnić, "Prediction-powered inference," *Science*, vol. 382, no. 6671, pp. 669–674, 2023, doi: 10.1126/science.adi6000.

[10] T. Zrnić and E. J. Candès, "Cross-prediction-powered inference," *Proc. Natl. Acad. Sci. USA*, vol. 121, no. 15, Art. no. e2322083121, 2024, doi: 10.1073/pnas.2322083121.

[11] A. Fisch, J. Maynez, R. A. Hofer, B. Dhingra, A. Globerson, and W. W. Cohen, "Stratified prediction-powered inference for effective hybrid evaluation of language models," in *Advances in Neural Information Processing Systems*, 2024. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2024/hash/c9fcd02e6445c7dfbad6986abee53d0d-Abstract-Conference.html.

[12] A. N. Angelopoulos, J. C. Duchi, and T. Zrnić, "PPI++: Efficient prediction-powered inference," 2024, arXiv:2311.01453v2. [Online]. Available: https://arxiv.org/abs/2311.01453v2.

[13] P. Mani, P. Xu, Z. C. Lipton, and M. Oberst, "No free lunch: Non-asymptotic analysis of prediction-powered inference," in *Proc. Int. Conf. Mach. Learn. (ICML)*, PMLR, vol. 306, 2026, pp. 86296–86335. [Online]. Available: https://proceedings.mlr.press/v306/mani26b.html.

[14] J. Kossen, S. Farquhar, Y. Gal, and T. Rainforth, "Active testing: Sample-efficient model evaluation," in *Proc. Int. Conf. Mach. Learn. (ICML)*, PMLR, vol. 139, 2021, pp. 5753–5763. [Online]. Available: https://proceedings.mlr.press/v139/kossen21a.html.

[15] F. Lin, N. Krever, J. Černý, and L. Xu, "Not all mislabels are equal: Efficient task-aware auditing with confidence sequences," 2026, arXiv:2610.33262. [Online]. Available: https://arxiv.org/abs/2610.33262.

[16] T.-Y. Lin et al., "Microsoft COCO: Common objects in context," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2014, pp. 740–755, doi: 10.1007/978-3-319-10602-1_48.

[17] M. Everingham, L. Van Gool, C. K. I. Williams, J. Winn, and A. Zisserman, "The PASCAL visual object classes (VOC) challenge," *Int. J. Comput. Vis.*, vol. 88, pp. 303–338, 2010, doi: 10.1007/s11263-009-0275-4.

[18] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Advances in Neural Information Processing Systems*, vol. 28, 2015. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2015/hash/14bfa6bb14875e45bba028a21ed38046-Abstract.html.

[19] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, 2017. [Online]. Available: https://arxiv.org/abs/1708.02002.

[20] TorchVision contributors, "Faster R-CNN and RetinaNet implementations and COCO weights," version 0.23.0. Accessed: Nov. 11, 2026. [Online]. Available: https://github.com/pytorch/vision/tree/v0.23.0/torchvision/models/detection.